College of Engineering – Department of Chemical Engineering

CURRICULUM

Bachelor of Science in Food Engineering
Academic Year 2025-2026
Reference CMOs: CMO No. 4 s. 2018 and CMO No. 20, s. 2013, CMO No. 39 s. 2021

Curriculum Description

Food Engineering is a multidisciplinary field of applied physical sciences which combines science, microbiology, and engineering education for food and related industries. Food engineering includes, but is not limited to, the application of agricultural engineering and chemical engineering principles to food materials. Food engineers provide the technological knowledge transfer essential to the costeffective production and commercialization of food products and services

Program Education Objectives

The food engineering alumni three to five years after graduation shall:
PEO 1: Successfully practice as defined in the law governing the scope of Food Engineering.
PEO 2: Assume leadership position in industry, academe, government or private sector with consideration to social and ethical responsibility.
PEO 3: Engage in lifelong learning through further studies, research, certifications, promotions, and other personal and professional development activities.

Student Outcomes

The following skills, knowledge, and values are expected to be attained by students as they progress through the program:
SO1 Discipline Knowledge. Ability to apply mathematics, sciences and principles of engineering to solve complex engineering problems;
SO2 Investigation. Ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions;
SO3 Design/Development of Solutions. Design solution, system, components, processes, exhibiting improvements/innovations, that meet specified needs with appropriate consideration for public health and safety, cultural, societal, economical, ethical, environmental and sustainability issues.
SO4 Leadership and Teamwork. Function effectively as a member of a leader on a diverse team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.
SO5 Problem Analysis. Identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics;
SO6 Ethics and Professionalism. Apply ethical principles and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, environmental, and societal contexts.
SO7 Communication. Communicate effectively on complex engineering activities with the community, and the society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions;
SO8 Environment and Sustainability. Recognize the impact of professional engineering solutions in societal, global, and environmental contexts and demonstrate knowledge of and need for sustainable development;

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SO9 Lifelong Learning. Recognize the need for, and ability to engage in independent and life-long learning in the broadest context of technological change.

SO10 The Engineer and Society. Apply reasoning based on contextual knowledge to assess societal, health, safety, legal, cultural, contemporary issues, and the consequent responsibilities relevant to professional engineering practices.

SO11 Modern Tool Usage. Apply appropriate techniques, skills, and modern engineering and IT tools to complex engineering activities.

SO12 Project Management and Finance. Demonstrate knowledge and understanding of engineering management and financial principles as member or a leader of a team to manage projects in multidisciplinary settings and identify opportunities of entrepreneurship.

SO13 Social and National Responsibility. Apply acquired engineering knowledge and skills in addressing community problems that contributes to national development.

CURRICULUM COMPONENTS

Classification / Field / Course	Number of Hours per Week		Credit Units
	Lec	Lab	
I. TECHNICAL COURSES			
A. Mathematics			
MATH 101 - Differential Calculus	4	0	4
MATH 102 - Integral Calculus	4	0	4
MATH 103 -Differential Equations	3	0	3
AI 102 - Probability and Statistics for AI	3	0	3
Subtotal	**14**	**0**	**14**
B. Natural/ Physical Sciences			
CHEM 101 - Chemistry for Engineers	3	3	4
BioT 101 - Biotechnology	3	0	3
PHYS 101 -Physics for Engineers	3	3	4
Subtotal	**9**	**6**	**11**
C. Basic Engineering Sciences			
ENGG 101 - Introduction to Engineering	0	3	1
ENGG 102 - Engineering Drawing	0	3	1
ENGG 103 - Computer-Aided Design	0	3	1
ENGG 104 - Engineering Economics	3	0	3
ENGG 106 - Engineering Management	2	0	2
ENGG 113 - Environmental Science and Engineering	3	0	3
ENGG 118 - Materials Science and Engineering	3	0	3
CpE 101 - Computer Programming	0	3	1
Subtotal	**11**	**12**	**15**
D. Allied Courses			
CHEM 201 - Analytical Chemistry	4	3	5
CHEM 202 - Organic Chemistry	4	3	5
CHE 208 - Physical Chemistry for Engineers	2	3	3
CHE 310 - Industrial Waste Management and Control	3	0	3
CHE 204 - Chemical Engineering Thermodynamics	2	3	3
EE 213 - Basic Electrical and Electronics	2	3	3

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Engineering			
ENGG 105 - Technopreneurship	3	0	3
ENGG 109 - Engineering Mechanics	3	0	3
Subtotal	**21**	**12**	**28**
E. Professional Courses			
FE 201 - Food Engineering Calculations	2	3	3
FE 202 - Nutrition Science	2	0	2
FE 203 - Advanced Engineering Mathematics for FE	3	0	3
FE 204 - Flow of Fluids	2	3	3
FE 205 - Food Chemistry	2	3	3
FE 206 - General Microbiology	2	3	3
FE 301 - Heat and Mass Transfer in Food	2	3	3
FE 302 - Food Microbiology	2	3	3
FE 303 - Computer Applications in Food Engineering	0	3	1
FE 304 - Separation in Food Processes	2	3	3
FE 305 - Sensory Evaluation	2	3	3
FE 306 - Food Research and Development	0	3	1
FE 307 - Food Processing	2	3	3
FE 308 - Technical Analysis of Food	2	3	3
FE 309 - Food Engineering Project 1	0	3	1
FE 310 - Food Processing Industries	3	0	3
FE 311 - Plant Visit and Seminars	0	3	1
FE 401 - Post-Harvest Handling Technology	3	0	3
FE 402 - Food Packaging and Labeling	3	0	3
FE 403 - Biochemistry for Food Engineers	3	0	3
FE 404 - Laws, Ethics and Process Safety for Food Engineering	3	0	3
FE 405 - Food Engineering Design 1	1	3	2
FE 406 - Food Engineering Project 2	0	3	1
FE 407 - Food Engineering Design 2	0	6	2
FE 408 - Food Engineering Integrative Studies	0	6	2
CHE 303 - Process Dynamics and Control	2	3	3
ENGG 116 - Methods of Research	3	0	3
ENGG 117 - OJT/Internship (320 hrs)	320		4
FEE 301 - Elective 1	3	0	3
FEE 401 - Elective 2	3	0	3
Subtotal	**52**	**383**	**77**
TOTAL TECHNICAL COURSES	**107**	**413**	**145**
II. NON-TECHNICAL COURSES			
A. General Education Courses			
GEd 101 - Understanding the Self	3	0	3
GEd 102 - Mathematics for the Modern World	3	0	3
GEd 104 - Contemporary World	3	0	3
GEd 105 - Readings in Philippine History	3	0	3
GEd 106 - Purposive Communication	3	0	3
GEd 107 - Ethics	3	0	3

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GEd 108 - Art Appreciation	3	0	3
GEd 109 - Science, Technology, and Society	3	0	3
Subtotal	**24**	**0**	**24**
B. General Education Electives			
AI 101 - Linear Algebra for AI	3	0	3
AI 103 - Machine Learning and Neural Networks	2	3	3
GEdE 101 - Entrepreneurial mind	3	0	3
Subtotal	**8**	**3**	**9**
C. Mandated Courses			
GEd 103 - Life's and Works of Rizal	3	0	3
PATHFit 1 - Movement Competency Training	2	0	2
PATHFit 2 - Exercise-based Fitness Activities	2	0	2
PATHFit 3 - Choice of Dance, Sports, Martial Arts, Group Exercise, Outdoor, & Adventure Activities	2	0	2
PATHFit 4 - Choice of Dance, Sports, Martial Arts, Group Exercise, Outdoor, & Adventure Activities	2	0	2
NSTP 111 - National Service Training Program 1	3	0	3
NSTP 121 - National Service Training Program 2	3	0	3
Subtotal	**17**	**0**	**17**
TOTAL NON-TECHNICAL COURSES	**49**	**3**	**50**
GRAND TOTAL	**156**	**416**	**195**

SUMMARY

Courses	Number of Units
I. TECHNICAL COURSES	
A. Mathematics	**14**
B. Natural/Physical Sciences	**11**
C. Basic Engineering Sciences	**15**
D. Allied Courses	**28**
E. Professional Courses	**77**
II. NON-TECHNICAL COURSES	
A. General Education Courses	**24**
B. General Education Electives	**9**
C. Mandate Courses	**17**
GRAND TOTAL	**195**

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CURRICULUM STRUCTURE

FIRST YEAR							
FIRST SEMESTER							
Course code	Course Title	Credit Units	No of Hr./s		Pre-requisite(s)	Co-requisites	Category
			Lec	Lab			
ENGG 101	Introduction to Engineering	1	0	3			TC - C
GEd 105	Readings in Philippine History	3	3	0			NTC - A
GEd 104	The Contemporary World	3	3	0			NTC - A
GEd 108	Art Appreciation	3	3	0			NTC - A
MATH 101	Differential Calculus	4	4	0			TC - A
NSTP 111	National Service Training Program 1	3	3	0			NTC - D
PATHFit 1	Movement Competency Training	2	2	0			NTC - C
CHEM 101	Chemistry for Engineers	4	3	3			TC - B
GEd 102	Mathematics for the Modern World	3	3	0			NTC - A
ENGG 102	Engineering Drawing	1	0	3			TC - C
	Total	**27**	**24**	**9**			

FIRST YEAR							
SECOND SEMESTER							
Course code	Course Title	Credit Units	No of Hr./s		Pre-requisite(s)	Co-requisites	Category
			Lec	Lab			
CpE 101	Computer Programming	1	0	3			TC - C
GEd 101	Understanding the Self	3	3	0			NTC - A
GEd 106	Purposive Communication	3	3	0			NTC - A
GEd 109	Science, Technology and Society	3	3	0	ENGG 101		NTC - A
MATH 102	Integral Calculus	4	4	0	MATH 101		TC - A
PHYS 101	Physics for Engineers	4	3	3	MATH 101	MATH 102	TC - B
NSTP 121	National Service Training Program 2	3	3	0	NSTP 111		NTC - D
PATHFit 2	Exercise-based Fitness Activities	2	2	0	PATHFit 1		NTC - C
GedE 101	Entrepreneurial mind	3	3	0			NTC - B
	Total	**26**	**24**	**6**			

FIRST YEAR							
MIDTERM							
Course code	Course Title	Credit Units	No of Hr./s		Pre-requisite(s)	Co-requisites	Category
			Lec	Lab			
GEd 103	Life and Works of Rizal	3	3	0			NTC - B
BioT 101	Biotechnology	3	3	0			TC - B
GEd 107	Ethics	3	3	0			NTC - A
	Total	**9**	**9**	**0**			

SECOND YEAR
FIRST SEMESTER

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Course code	Course Title	Credit Units	No of Hr./s		Pre-requisite(s)	Co-requisites	Category
			Lec	Lab			
CHEM 201	Analytical Chemistry	5	4	3	CHEM 101		TC - D
CHEM 202	Organic Chemistry	5	4	3	CHEM 101		TC - D
FE 201	Food Engineering Calculations	3	2	3	CHEM 101, PHYS 101		TC - E
FE 202	Nutrition Science	2	2	0	BioT 101		TC - E
MATH 103	Differential Equations	3	3	0	MATH 102		TC - A
AI 101	Linear Algebra for AI	3	3	0			NTC - B
ENGG 109	Engineering Mechanics	3	3	0	PHYS 101		TC - D
PATHFit 3	Choice of Dance, Sports, Martial Arts, Group Exercise, Outdoor, & Adventure Activities	2	2	0	PATHFit 2		NTC - C
	Total	**26**	**23**	**9**			

SECOND YEAR							
SECOND SEMESTER							
Course code	Course Title	Credit Units	No of Hr./s		Pre-requisite(s)	Co-requisites	Category
			Lec	Lab			
CHE 208	Physical Chemistry for Engineers	3	2	3	MATH 103, CHEM 101		TC - D
FE 203	Advanced Engineering Mathematics for FE	3	3	0	MATH 201		TC - E
FE 204	Flow of Fluids	3	2	3	FE 201, MATH 103		TC - E
FE 205	Food Chemistry	3	2	3	CHEM 201, CHEM 202		TC - E
FE 206	General Microbiology	3	2	3	BioT 101		TC - E
ENGG 103	Computer-Aided Design	1	0	3	ENGG 102		TC - C
AI 102	Probability and Statistics for AI	3	3	0	MATH 102		TC - A
ENGG 113	Environmental Science and Engineering	3	3	0	CHEM 101		TC - C
PATHFit 4	Choice of Dance, Sports, Martial Arts, Group Exercise, Outdoor, & Adventure Activities	2	2	0	PATHFit 3		NTC - C
	Total	**24**	**19**	**15**			

THIRD YEAR							
FIRST SEMESTER							
Course code	Course Title	Credit Units	No of Hr./s		Pre-requisite(s)	Co-requisites	Category
			Lec	Lab			
FE 301	Heat and Mass Transfer in Food	3	2	3	FE 204		TC - E
FE 302	Food Microbiology	3	2	3	FE 206		TC - E
FE 303	Computer Applications in Food Engineering	1	0	3	CpE 101		TC - E
CHE 303	Process Dynamics and Control	3	2	3	FE 203, ENGG 103		TC - E
CHE 204	Chemical Engineering Thermodynamics	3	2	3	PHYS 101, MATH 102		TC - D

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ENGG 104	Engineering Economics	3	3	0	MATH 101		TC - C
ENGG 116	Methods of Research	3	3	0	GEd 106,CHE 208, AI 102, FE 201		TC - E
ENGG 118	Materials Science and Engineering	3	3	0	CHEM 101		TC - C
EE 213	Basic Electrical and Electronics Engineering	3	2	3	PHYS 101		TC - D
	Total	**25**	**19**	**18**			

THIRD YEAR							
SECOND SEMESTER							
Course code	**Course Title**	**Credit Units**	**No of Hr./s**		**Pre-requisite(s)**	**Co-requisites**	**Category**
			Lec	**Lab**			
FE 304	Separation in Food Processes	3	2	3	CHE 204, FE 301		TC - E
FE 305	Sensory Evaluation	3	2	3	AI 102, FE 205		TC - E
FE 306	Food Research and Development	1	0	3	FE 202, FE 205, FE 302		TC - E
FE 307	Food Processing	3	2	3	FE 202, FE 205		TC - E
FE 308	Technical Analysis of Foods	3	2	3	FE 205		TC - E
FE 309	Food Engineering Project 1	1	0	3	ENGG 116		TC - E
CHE 310	Industrial Waste Management and Control	3	3	0	ENGG 113		TC - D
AI 103	Machine Learning and Neural Networks	3	2	3	AI 101, AI 102, CpE 101		NTC - B
FEE 301	Elective 1	3	3	0	Third Year Standing		TC - E
	Total	**23**	**16**	**21**			

THIRD YEAR							
MIDTERM							
Course code	**Course Title**	**Credit Units**	**No of Hr./s**		**Pre-requisite(s)**	**Co-requisites**	**Category**
			Lec	**Lab**			
FE 310	Food Processing Industries	3	3	0			TC - E
FE 311	Plant Visit and Seminars	1	0	3			TC - E
ENGG 106	Engineering Management	2	2	0	ENGG 104		TC - C
	Total	**6**	**5**	**3**			

FOURTH YEAR							
FIRST SEMESTER							
Course code	**Course Title**	**Credit Units**	**No of Hr./s**		**Pre-requisite(s)**	**Co-requisites**	**Category**
			Lec	**Lab**			
FE 401	Post Harvest Handling Technology	3	3	0	FE 307, FE 310		TC - E
FE 402	Food Packaging and Labeling	3	3	0	FE 202, ENGG 118		TC - E
FE 403	Biochemistry for Food Engineers	3	3	0	CHEM 201, CHEM 202		TC - E
FE 404	Laws, Ethics and Process Safety for Food Engineering	3	3	0	GEd 107, FE 310		TC - E

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FE 405	Food Engineering Design 1	2	1	3	FE 304, FE 310, CHE 303		TC - E
FE 406	Food Engineering Project 2	1	0	3	FE 309, AI 102		TC - E
ENGG 105	Technopreneurship	3	3	0	Fourth Year Standing		TC - D
FEE 401	Elective 2	3	3	0	Fourth Year Standing		TC - E
	Total	**21**	**19**	**6**			

FOURTH YEAR							
SECOND SEMESTER							
Course code	**Course Title**	**Credit Units**	**No of Hr./s**		**Pre-requisite(s)**	**Co-requisites**	**Category**
			Lec	**Lab**			
FE 407	Food Engineering Design 2	2	0	6	FE 405		TC - E
ENGG 117	On the Job Training	4	320		Fourth Year Standing		TC - E
FE 408	Food Engineering Integrative Studies	2	0	6	Fourth Year Standing		TC - E
	Total	**8**	**0**	**332**			